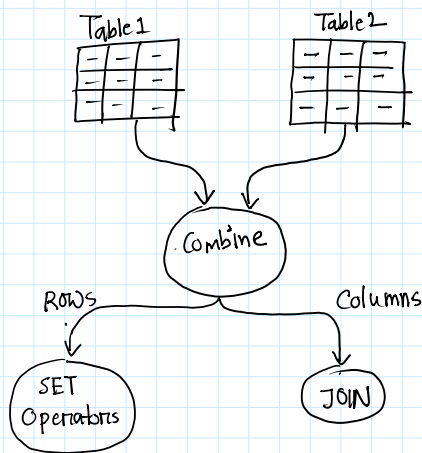


Methods for combining 2 tables



JOIN

say,
1st Table → Table A

-	-	-
-	-	-
-	-	-

(Requires Key columns)

2nd Table → Table B

-	-	-
-	-	-
-	-	-

Joining Tables A & B gives,

-	-	-	-	-
-	-	-	-	-
-	-	-	-	-

Table A on
the left

Table B on
the right

The columns have been joined.

SET operators

Table A and B have same rows.

-	-	-
-	-	-
-	-	-

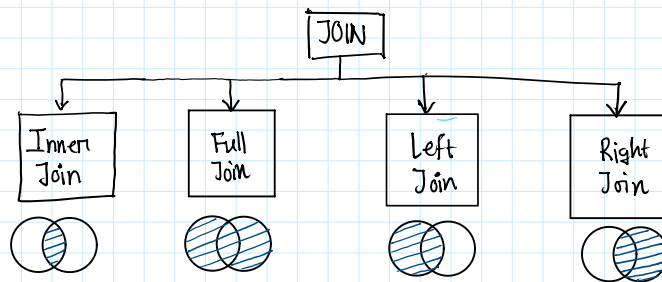
-	-	-
-	-	-
-	-	-

(Column numbers
must be the same).

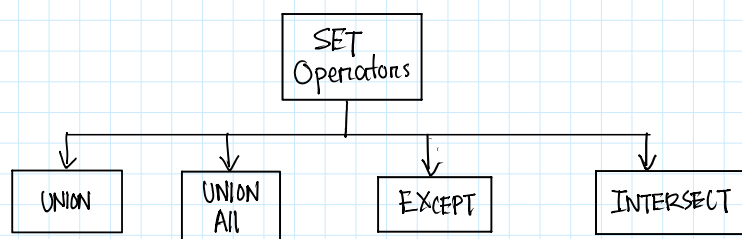
Append B to A using
SET operators.

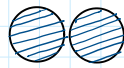
-	-	-
-	-	-
-	-	-
-	-	-
-	-	-

Types of JOIN



Types of SET Operators





JOIN

id	name
1	A
2	B
3	C

id	country
1	D
2	E
4	F

→ 'id' is common column in both table.

Need to join these 2 using id

Types



INNER



LEFT-ANTI



FULL



RIGHT-ANTI



LEFT



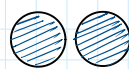
FULL-ANTI



RIGHT



CROSS



NO JOIN

Advanced

Basics

Why join?

1. Recombine Data
 - put data together from spreaded sources (tables)
2. Data enrichment
 - Get extra information from other sources
3. Check for Existence
 - Check for data existence in other sources.

NO JOIN

Syn: SELECT * FROM tableA;
SELECT * FROM tableB;

Make separate queries.
No data joining.

INNER JOIN

- Returns only matching rows from both tables.

Data from table A and table B are matching.
See those data.

Syntax:

SELECT *
FROM A
[join-type] JOIN B
ON <condition>

SELECT *
FROM A
INNER JOIN B
ON A.key = B.key

Join using selected columns

SELECT
col-1
A, col-2 B

ON <condition>

ON A.key = B.key

Purpose: Recombining data

```
SELECT  
col-1  
A.col-2A  
A.col-3A  
B.col-4B  
col-5  
FROM A  
[type] JOIN B  
ON A.col = B.col
```

LEFT JOIN

All data from the left table.
but

Only matching data from right table.

If data does not exist,
We will get NULL

→ Order of the table is very important

Syntax:

```
SELECT
```

```
=====
```

```
FROM <left-table> as lt
```

```
LEFT JOIN <right-table> as rt
```

```
ON _____ = _____
```

RIGHT JOIN

Return all rows from right table
but

only matching data from left table

→ Order is important.

Syntax:

```
SELECT
```

```
=====
```

```
FROM <left-table>
```

```
RIGHT JOIN <right-table>
```

```
ON _____ = _____
```

The exact opposite of LEFT JOIN

LEFT JOIN is more popular

FULL JOIN

Return all the rows from both tables

Syntax:

```
SELECT
```

```
=====
```

Syntax:

SELECT

FROM <A-table> As A

FULL JOIN <b-table> As B

ON A.~~key~~ = B.~~key~~